

HOMework 10

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 12/06/2023

Problem 1. Consider the ODE

$$u'(t) = \lambda(u(t) - \cos(t)) - \sin(t).$$

- (a) Verify that one particular solution is $u(t) = \cos(t)$. If we consider the initial data of the form $u(t_0) = \eta$ that does not lie on the curve of $\cos(t)$, then the solution through this point is a different function. Verify that the function is

$$u(t) = e^{\lambda(t-t_0)}(\eta - \cos(t_0)) + \cos(t).$$

Supposing that $\operatorname{Re}(\lambda) < 0$, note that this function approaches $\cos(t)$ exponentially quickly with decay rate λ .

- (b) Consider the problem with $\lambda = -10^6$. Implement the trapezoidal and backward Euler methods for the following two cases for the final time $T = 3$ by using various temporal step sizes $h = 0.1, 0.2$, and 0.5 . Then, given the cases listed below, do the following:
- (i) Make a table to show the errors for the two cases by using the two methods with the various values of h (columns for the methods and rows for the two cases with various h).
 - (ii) Plot your numerical solutions against the true solution in Case 2 and show the comparison of the trapezoidal method and the backward Euler method.

Case 1: Take the initial data $u(0) = 1$, so that $u(t) = \cos(t)$ is the solution.

Case 2: Take the initial data $u(0) = 1.5$, so that there is an initial rapid transient toward $u = \cos(t)$ on a time scale about 10^{-6} .